OO-S-00256/16B(GL)
October 7, 1988
INTERIM REVISION OF
OO-S-256/16A
June 13, 1986

INTERIM FEDERAL SPECIFICATION SHEET

SEWING MACHINE, INDUSTRIAL; DARNING, CYLINDER BED,

SINGLE NEEDLE, 301 STITCH, PARACHUTE REPAIR

This interim Federal specification was developed by the U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations of changes to the preparing activity at the address shown above.

The General Services Administration has authorized the use of this Interim Federal Specification as a valid exception to OO-S-256/16A, dated June 13, 1986.

The complete requirements for procuring the sewing machines described herein shall consist of this document and the issue in effect of Specification OO-S-256.

1. SCOPE

1.1. Scope. This specification covers the specific requirements for single needle, flatbed, 301 stitch, heavy duty, darning, industrial sewing machines used in the fabrication and repair of aerial delivery equipment.

2. APPLICABLE DOCUMENTS

2.1 Government publications. The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY NATICK RESEARCH AND DEVELOPMENT CENTER

6-1-9367 Sewing Machine, Industrial, Darning, Single Needle, Heavy Duty: Parachute Repair and Fabrication Test Requirements

FSC 3530

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:59Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Machine, heavy duty. The heavy duty darning machine shall have the capability of darning light to heavy cotton and nylon duck utilizing metric needle sizes 90, 110, 120, 125, 130, 140, and 180. The machine shall operate at speeds not less than 2000 or greater than 3000 stitches per minute. The machine shall be manually or semi-automatically lubricated. The machine shall be furnished with a cone on the end of the cylinder bed. The machine shall be pedestal mounted. Each machine shall be supplied with a bobbin winder and knee lifter.

3.1.1 Machine feed. The machine shall not have a type of feeding mechanism.

3.1.2 Machine height under the presser foot. The machine shall provide a minimum height of 3/8 of an inch under the presser foot.

3.1.3 Machine, working room to the right of the needle. The machine shall be furnished with a working room distance to the right of the needle of 10 inches minimum to 16 inches maximum.

3.1.4 Machine, table top. The machine shall be furnished with a type II table top. The outside dimensions of the table top shall be 20 by 40 by 2 inches thick with a tolerance of ±1/4 of an inch for each dimension.

3.1.5 Spare parts. The machine shall be furnished with the following spare parts in the quantity indicated.

Spare parts

Item Identification	Quantity
Face plate screws	2
Thread take up spring	1
Thread tension spring thumb nut	1
Tension stud	1
Sewing hook assembly	1
Sewing hook holding screw	1
Bobbin, metal	10
Bobbin case assembly	1
Saddle gear screw	1
Needle bar thread guide	1
Needle bar thread guide screw	2
Needle set screw	2
Lifting bracket screw	2
Presser foot finger guard spring	1
Presser foot holding screw	1
Presser foot	1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:59Z
Check the source to verify that this is the current version before use.

Spare parts (cont'd)

Item Identification	Quantity
Presser foot finger guard spring screw	2
Thread guide screw	2
Needle brushing thread guide	1
Top plate holding screw	1
Take up lever stud	1
Needle, round point, Chrome finish:	
Metric sizes:	
90	100
110	100
125	100
140	100
180	100

3.2 Performance.

3.2.1 Machine operation. The heavy duty machine shall be capable of darning and mending panels, pack sections and deployment bags of personnel, extraction and cargo parachutes when tested as specified in 4.1.1 and 4.2.1. The machine shall show no signs of stalling, pounding, shearing of threads, failure to lock stitches of deformation of any part during the darning and mending of these articles.

4. QUALITY ASSURANCE PROVISIONS

4.1 First article testing. When a first article is required it shall be tested in accordance with the applicable test method specified in table I.

TABLE I. First article tests

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.1.1

4.1.1 Machine test. The heavy duty machine shall be operated to darn and mend all of the typical parachute details as shown on Drawing 6-1-9367. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test.

4.2 End item testing. Each end item shall be tested in accordance with the applicable test method specified in table II. Failure of any test shall be cause for rejection of the end item involved.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:59Z
Check the source to verify that this is the current version before use.

TABLE II. End item testing

Sewing machine	Requirement paragraph	Test method
Heavy duty	3.2.1	4.2.1

4.2.1 Machine test. Each machine shall be operated to sew samples of typical parachute construction, as shown by end item test detail on Drawing 6-1-9367. Any nonconformance with the requirements of 3.2.1 shall constitute failure of this test. The completed test samples shall be supplied with each machine furnished.

5. PACKAGING

This section does not apply to this detailed specification.

6. NOTES

6.1 Intended use. The sewing machines are intended for industrial use for fabricating and repairing parachutes.

6.2 Ordering data.

Title, number and date of this specification.

Custodian:

Army - GL

Preparing activity:

Army - GL

Project No. 3530-A086

★U.S. GOVERNMENT PRINTING OFFICE: 1989 - 604-033/11372

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T11:59Z
Check the source to verify that this is the current version before use.